$u_c(C_p) = 0.02993$ mg/kg as indicated in the bottom line of Table 6.3. The expanded 95% uncertainty is obtained by multiplying it by 2, which gives:

$$U(Z)_{95\%} = 2 \times 0.02993 = 0.05986\ \text{mg/kg}$$

And the relative uncertainty:

$$UR\%(\text{Z}) = \frac{0.05986}{1.99998} \times 100 = 2.99\%$$

The result can then be expressed as:

$$Z = 1.99998\ (\pm 0.05986)\ \text{mg/kg}$$

This means that 95% of the true values, or considered as true values of C_p, are within the coverage interval [1.94012, 2.05984] mg/kg.

In most cases, the coverage probability is conventionally chosen to be 95%, i.e. $k_{GUM} = 2$. When the method of estimating the standard uncertainty makes it easy to obtain the real number of df, it is preferable to take the exact coverage factor, based on this number of df. In Section 7.2 the MU estimation based on the β-ETI is explained and the parameter N_E, called the number of effective measurements, is used to obtain the exact coverage factor and compared to the standardized.

6.9 Round the Result

The rounding of results is a very classic practice yet required by regulatory bodies. It is based on the concept of the number of significant figures. Above all, it is a simple and intuitive way of indicating the level of uncertainty that the experimenter gives to the measurements, and it is also a method to simplify their reading. Therefore, rounding can be considered as a method to express uncertainty. To perform result rounding it is thus possible to use its MU.

In some contexts, rounding is a regulatory requirement. For instance, the United States Pharmacopeia (USP) defined a standard operating procedure (SOP) for rounding analytical results. In this context, three different procedures are proposed and considered in agreement with three types of measurements: dissolution results, impurity and results close to limit of detection (LOD), and other results.

While using computers, it may seem superfluous to round results as far as the internal machine representation of a data contains a considerable number of figures. For example, an Excel worksheet uses a 64-bit floating-point internal format and generally stores data in 32 bits or more, though it is converted into an internal memory format, the number of significant digits can be extremely high. Above all, the data, such as it appears on the screen is not rounded in the sense that it is understood here but depends on the display format chosen by the user, whereas the calculations are done on all 32 bits or more.

The main disadvantage of rounding is that it necessarily generates a bias. It is therefore imperative to apply rounding only after all intermediate calculations have been completed. Many theoretical approaches have been published [8] but, according to ISO 17025 standard, the choice of the rounding method and the number of significant digits is left to the laboratory, which can thus define its own rules [11]. Two simple rounding rules are applicable:

- When the first significant digit is between 5 and 9, the result is rounded to this decimal place and the MU will thus include one significant digit.
- When the first significant digit is <5, the result will be rounded to that next decimal place and the MU will therefore include two significant digits.

These rules can be translated into a mathematical formula, shown in Eq. (6.23). Let *Dn* be a decimal number with *ns* significant digits and **int**{ } the operator for obtaining the integer part of a decimal number. The following formula allows to determine *ns* by passing through the decimal logarithm of the number which is used to isolate the number of significant figures. The rounding is then done by considering *ns*.

$$ns = \mathbf{int}\{-\mathrm{Log}_{10}(Dn) + 1\} \tag{6.23}$$

Resource L Rounding a result (Excel).

	A	B	C	D	E	F
1	**Resource L: Rounding a result**					
2		Raw data	Rounded data	Formula		
3	Value	1.999982505	2.00	=ROUND(B3;INT(-LOG10(B4))+1)		
4	Uncertainty	0.059869686	0.06	=ROUND(B4;INT(-LOG10(B4))+1)		
5						
6	**Examples**					
7	Z	U(Z)	Siginificant figures	Z*	U(Z)*	
8	1.99998251	0.059869686	2	2.000	0.060	
9	75.23678	0.0278	2	75.240	0.030	
10	75.23678	0.00568	3	75.237	0.006	
11	75.23678	12.3689	–1	80.000	10.000	

The Resource L worksheet illustrates this formula in Excel and is applicable to data such as the LEAD example. The raw result is in column `A`, the expanded uncertainty in column `B`, the rounded values in column `C`, and the Excel formula in column `D`. First, two internal functions are used `LOG10`, which provides the decimal logarithm, and `INT` which retains only the integer part. Finally the built-in function, `ROUND` completes the calculation using as argument the output of two functions. Finally, the Lead result can be expressed in various equivalent forms:

$$Z = 2.00 \pm 0.06\ \mathrm{mg/kg}$$

$$Z = 2.00 \pm 3\%\ \mathrm{mg/kg}$$

$$Z = [1.94, 2.06]\ \mathrm{mg/kg}$$

6.10 Accuracy, Total Error, and Uncertainty

The introduction of the concepts of trueness, precision, and accuracy was mainly aiming to characterize a measuring method. Analysts are now familiar with these characteristics, although definitions may be fluctuating. On the other hand, MU is

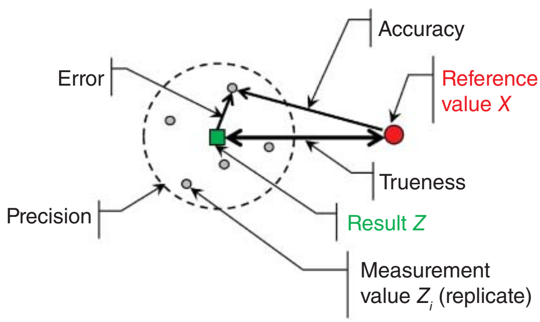

Figure 6.5 Schematic representation of the main concepts used for method validation.

no longer focuses on the measurment method but on the analytical result, which can be described as the combination of the method, the laboratory, the type of matrix, the operator on a given day, and so on. To be explicit, the purpose of an interlaboratory analysis is to characterize the analytical method, using its standard deviation of reproducibility; it is not directly aimed to qualify the analytical result issued by a laboratory.

Figure 6.5 provides the schematic illustration of accuracy and the various characteristics or parameters used for method validation. According to the VIM definitions, trueness is the closeness of agreement between the average of an infinite number of replicate measurements and the reference value; it is a distance. The term "closeness of agreement" was preferred to the term "difference" to indicate this is a concept while several parameters are applicable to estimate trueness. In the figure, trueness is symbolized by a double-headed arrow.

Similarly, precision is defined as the closeness of agreement between replicate measures; it characterizes a dispersion. It is represented as a circle containing a known proportion of measurements (without going into the details of the calculation).

The accuracy, on the other hand, characterizes the closeness of agreement of one individual measurement to the reference value. Because these concepts are represented as vectors on the figure, it is easy to see that accuracy, as defined in the VIM, is the combination of trueness and precision.

MU is also estimated with diverse standard deviations or standard uncertainties, like the precision parameters. But the rationale is different and more complicated, as multiple sources of variation are accounted for, like in a jigsaw puzzle. The solution proposed by metrologists is to combine the various standard deviations corresponding to these sources.

It can be tedious, and so we propose something slightly different. It is also possible to use an intermediate precision standard deviation to cover most of the sources of uncertainty. This last remark highlights the fundamental difference between a reproducibility standard deviation and a combined standard uncertainty. Uncertainty can contain a standard deviation of precision but not the opposite.

A classic confusion consists in thinking that repeatability or reproducibility constitutes the uncertainty of measurement [12]. Some analysts believed that the relative standard deviation of repeatability (RSD_r) is the appropriate estimate of relative uncertainty $UR\%$. However, RSD_r presents several major failings:

- It does not consider all the sources of variation included in the MU; therefore, it varies from day to day or from operator to operator.
- It does not consider the standard deviations of the measuring bias.
- It does not consider any coverage probability. This point is important as analysts often use the RSD_r to characterize a method. In its usual expression, it corresponds to 1 standard deviation related to a given average value. At best, the coverage probability is 67% since the interval covers ±1 standard deviation. It would have to be associated with a coverage factor of 2 for the proportion covered to be 95%. For instance, when the RSD_r is ±20% it means that 95% of the measurements are at ±40% around the announced result.

As already explained at the beginning of this chapter, in the field of clinical biology, the *TAE* has had some success. Like accuracy, it is based on a combination of trueness, using a bias, and precision introduced in the form of a standard deviation multiplied by a coverage factor to account the dispersion of measurements (formula 6.1) [13]. Figure 6.6 is a possible summarized comparison between *TAE* and MU. In the right part of the figure, the statements $\delta = 0$ and $u(\delta) \neq 0$ are not paradoxical but clearly underline two major principles of the GUM: the bias must be corrected before any MU estimation; and the bias standard uncertainty, if any, is included in the MU estimate.

In the conventional *TAE* model, for a given concentration level, the bias is considered as constant, this is the notion of systematic error, consequently its standard uncertainty is 0. Whereas in the MU model, the bias must be corrected and, because it is a random variable taking unpredictable values, it is possible to give an estimate of its standard uncertainty $u(\delta)$.

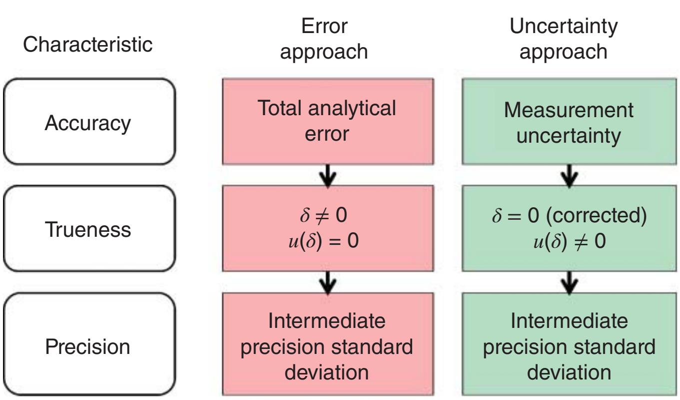

Figure 6.6 Comparison between the total analytical error (*TAE*) model and the measurement uncertainty (MU) model.

6.11 Insights on Probability

Estimating the MU is an operation whose objective is to evaluate the *probable* dispersion of the true values of a measurand. Given that statistics is the science of probability, estimating the MU requires a statistical approach. Indeed, modern statistics appeared at the beginning of the nineteenth century, at the same time as the scientific way of thinking was being established.

Initially, famous mathematicians such as Pierre de Fermat (1607–1665), Blaise Pascal (1623–1662) or Thomas Bayes (1702–1761) elaborated the theory of probability calculation because of the interest in games and the haphazard nature of winning. And it is not a coincidence, if the Arabic word *az-zhar* indicates the game of dice, while the origin of the word random is more confusing. The concept of probability derives from the notion of randomness and is mentioned many times in this book and is rather delicate to grasp because it is not always intuitive.

A classic starting point is to refer to the game of heads or tails and the experiment conducted by the British mathematician John Edmund Kerrich while interned in Nazi-occupied Denmark in the 1940s. It consisted of tossing a coin $n = 10{,}000$ times and counting the number of times it landed on heads; this number being denoted n_H. In one experiment, the result was $n_H = 5067$. From this experiment, the concept of probability of having heads, denoted Prob(H) can be defined, in a first meaning, as the relative frequency of the occurrence of heads: this leads to a possible definition of probability named frequential. That is, the ratio of the number of times heads was obtained to the total number of throws, denoted n, which gives:

$$\frac{n_H}{n} = \frac{5057}{10{,}000} = 0.5067 \text{ or } 50.67\%$$

This result is an observation of the real world. But intuitively it is easy to assume that the correct result should be equal to 0.5000 or 50.00% since a nonrigged or perfectly symmetrical coin has *theoretically* as much chance of falling on heads as on tails. By making this hypothesis, we unconsciously pass from the real world to an idealized world, posed *a priori*. To check if this passage is correct, it would require to be able to throw the coin an infinite number of times, which is of course unrealistic on the practical level, but which allows to propose a mathematical model, in the form:

$$\text{Prob}(H) = \lim_{n\to\infty} \frac{n_H}{n} = 0.5$$

It should also be noted that, when the experiment of 10,000-coin tosses is repeated, another value of n_H would be obtained different from the first one. The real world is therefore not easily enclosed in a simple mathematical formula. When the number of possible values taken by a measure is known in advance, as in the toss of a coin where there are two possible values or the choice of a card in a deck of 52 cards, the notion of probability defined as a relative frequency is simple to grasp.

Things become more complicated with measurements that can take an infinite (or almost infinite) number of values, as a concentration measurement value obtained with a quantitative analytical method. The contribution of mathematics to solve this